Cambridge (CIE) A Level Business

Your notes

The Nature of Operations

Contents

- The Transformational Process
- Efficiency, Productivity & Sustainability
- Methods of Operation (Production)

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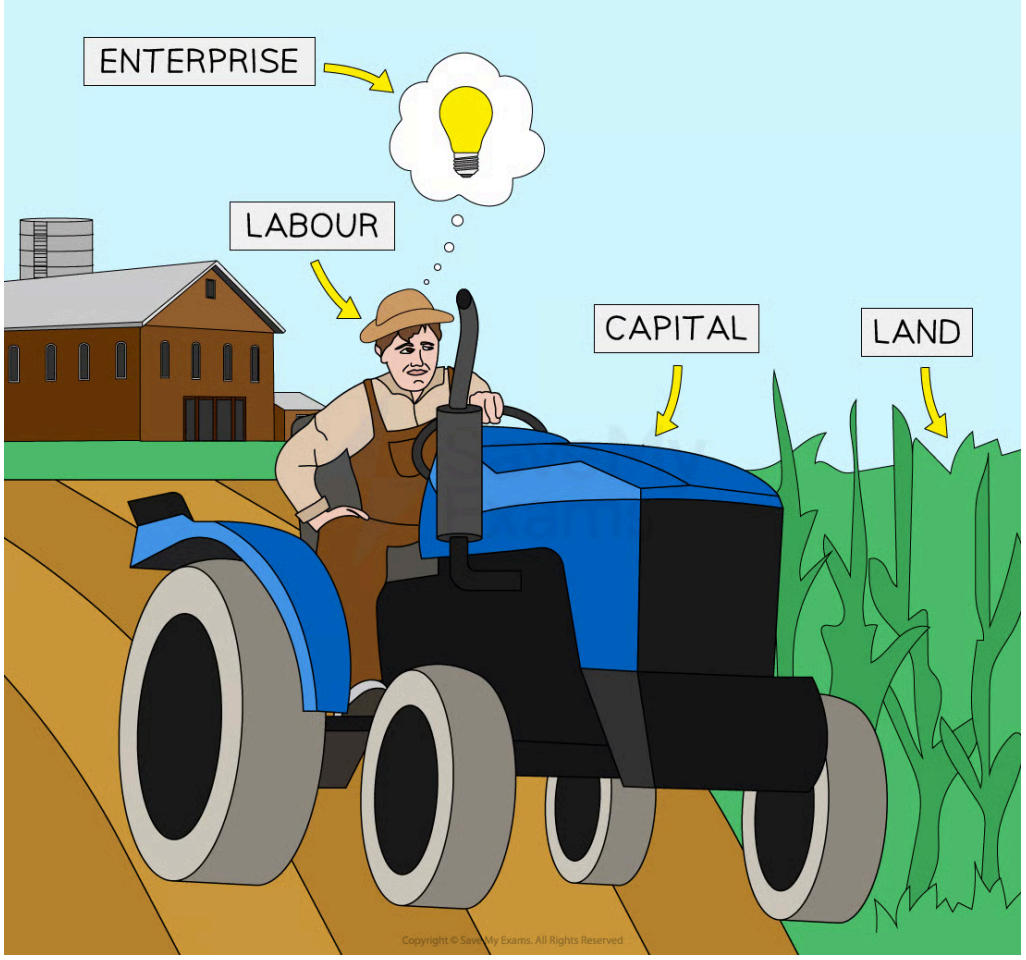
Using factors of production

- **Factors of production** are the resources used to produce goods and services



The four factors of production are land, capital, labour and enterprise

- The production of any good or service requires the use of **a combination of all four factors of production**

The four factors of production

1. Land

- **Non-man-made natural resources** available for production
- Some countries have a vast amount of a particular natural resource and are able to specialise in its production
  - E.g. **Kuwait** specialises in the product of oil, which accounts for 95% of its **exports**

2. Capital

- Any **man-made resource** that is used to produce goods or services
- Examples include tools, buildings, machines and computers

3. Labour

- The **human input** into the production process, labour involves mental or physical effort
- Not all labour is of the same quality
  - It can be **skilled or unskilled**
  - Some workers are more productive than others because of their education, training and experience

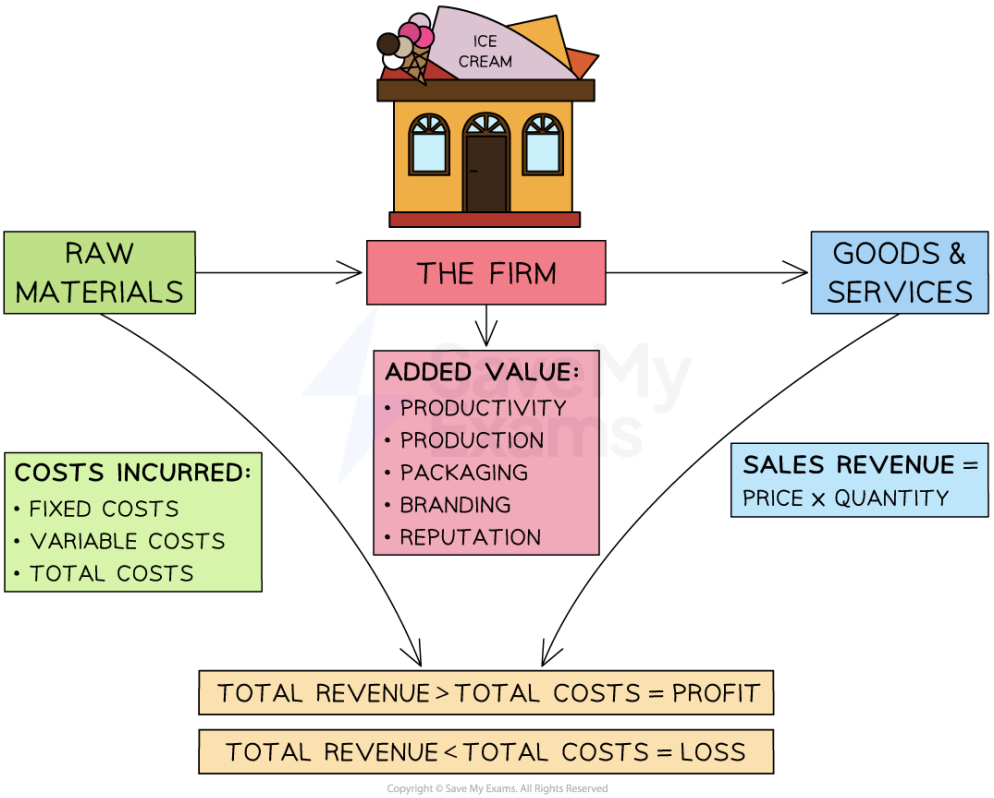
4. Enterprise

- Enterprise involves **taking risks** in setting up or running a firm
- An entrepreneur decides on the combination of the **factors of production** necessary to produce goods/services with the aim of generating profit

Stages of the transformational process

- Businesses take **inputs** and **transform** them in order to produce **outputs** that customers will want to buy

The transformational process



The transformational process

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- **Inputs** used in the transformation process may be financial, human or physical resources, as well as **enterprise**
- **Outputs** are goods and services, as well as **by-products** and waste
  - E.g. **Rothaus brewery** takes **inputs** including malts, hops and barley and uses the staff on the brewery premises in Freiburg, as well as equipment, such as mash tuns, to **transform** them by brewing these inputs into its **main output** — its range of beers — as well as **by-products**, such as compost

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Operations and added value

- Adding value is the process of **taking raw materials and using them in such a way that the end product created is worth more than the cost** of the raw materials used to create it - value has been added
  - E.g., customers pay more for potatoes when they are packaged as oven chips than they would be willing to pay for a bag of potatoes
  - If value is not added to the materials and components that a business buys, fixed costs cannot be paid and no profit will be made

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WAYS OF ADDING VALUE

CONVENIENCE

PRODUCTS WHICH OFFER A MORE CONVENIENT OPTION TO THE CUSTOMER CAN SELL FOR A HIGHER PRICE E.G. CHOPPED FRUIT & VEG SELLS FOR MORE THAN SINGLE FRUIT & VEG ITEMS

BRANDING

CREATING A UNIQUE IDENTITY THAT DISTINGUISHES IT FROM COMPETITORS INCREASES CUSTOMERS LOYALTY, TRUST, AND RECOGNITION WHICH TRANSLATES INTO HIGHER SALES AND PROFITABILITY

QUALITY

CUSTOMERS ARE WILLING TO PAY PREMIUM PRICES FOR PRODUCTS OR SERVICES THAT ARE OF HIGH QUALITY BECAUSE THEY OFFER GREATER VALUE AND SATISFACTION

DESIGN

DESIGN ADDS VALUE BY ENCHANCING THE AESTHETICS AND FUNCTIONALITY OF A PRODUCT/SERVICE. IT CAN ALSO IMPROVE THE USER EXPERIENCE AND INCREASE THE PERCIEVED VALUE OF THE PRODUCT/SERVICE

USP'S

BY IDENTIFYING A UNIQUE AND COMPELLING BENEFIT OR ADVANTAGE, COMPANIES CAN ATTRACT NEW CUSTOMERS, INCREASE SALES AND PROFITABILITY, AND BUILD A LOYAL CUSTOMER BASE

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Some of the methods of adding value allow for product differentiation, which allows the business to charge a higher selling price

Method	Example
Branding	▪ Apple has built a brand that many customers believe is **superior to other brands**

	▪ They use quality materials, innovative design and good marketing ▪ This branding allows Apple to **charge a higher price** for its products, increasing added value
Convenience	▪ Persil initially sold bottles of liquid for dishwashing machine use ▪ This resulted in spillage as customers added the liquid to their machines ▪ Persil developed **tablets**, which offer a more convenient option that can be sold at a higher price
Quality	▪ **Jo Malone** perfume products are well known for their **beautiful packaging,** which creates an exciting opening experience for the customer ▪ This allows the firm to charge a higher price for its products, increasing added value
Unique selling points (USPs)	▪ MoonPig birthday cards can be **completely customised** (size, colour, design etc) ▪ This customisation has helped Moonpig gain a competitive advantage ▪ It can charge a high price for its cards, increasing added value
Design	▪ **Samsung Galaxy Watch 5** has robust health tracking tools built into it, along with a highly advanced screen, ▪ These features allow the firm to **charge a higher price** for its products, increasing added value

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Efficiency, effectiveness, productivity and sustainability

- **Efficiency** means producing goods or services with the minimum amount of waste, time or resources
- **Effectiveness** means achieving the business's objectives by doing the right things to satisfy customer needs and meet goals
- **Productivity** is the volume of output produced per worker (or per hour, machine, etc.) that shows how well available resources are used to generate output

Why are these factors important?

Lower costs

- Efficient use of resources helps reduce waste and cut production costs

Higher profit margins

- Improved productivity and lower costs can increase the amount of profit made per sale

Better use of resources

- Efficient businesses make the most of time, money, staff and materials

Improved competitiveness

- Businesses that are productive and effective can offer competitive prices or higher-quality products

Customer satisfaction

- Being effective means meeting customer needs, which leads to loyalty and repeat purchases

Faster response to demand

- Productive operations allow quicker production and delivery, keeping up with customer demand

Supports business growth

- Efficient and productive businesses can expand more easily and take on larger orders

Measurement of labour productivity

- The **labour productivity** of a business is measure of the **output per worker** during a specified period of time
- Businesses aim to **increase the level** of labour productivity to **improve competitiveness**
- Labour productivity is calculated using the formula

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$$\text{Labour productivity} = \frac{\text{Output}}{\text{Number of workers}}$$

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Worked Example

The table shows the number of pairs of luxury wool socks produced by *Scotty Socks Ltd* in 2023 and 2024.

Year	Units produced
2023	46,000
2024	69,000

In 2023 *Scotty Socks* employed 50 staff. In 2024 the number of staff employed by the business increased by 20%.

Calculate the percentage change in labour productivity between 2023 and 2024.

(4)

Step 1: Calculate the labour productivity for 2023

$$= \frac{46{,}000 \text{ units}}{50 \text{ workers}} \quad \textbf{(1)}$$

$$= 920 \text{ units per worker}$$

Step 2: Calculate the labour productivity for 2024

$$= \frac{69{,}000 \text{ units}}{60 \text{ workers}} \quad \textbf{(1)}$$

$$= 1{,}150 \text{ units per worker}$$

Step 3: Calculate the percentage difference between the two years ((new-old) / old)

$$= \frac{1{,}150 - 920 \text{ units}}{920 \text{ units}} \times 100 \quad \textbf{(1)}$$

$$= 25\%$$

Step 4: Identify whether the percentage difference is an increase or decrease

- 25% percentage **increase** **(1)**

The link between high labour productivity and competitive edge

- Businesses aim to **increase the level** of labour productivity to **improve competitiveness**

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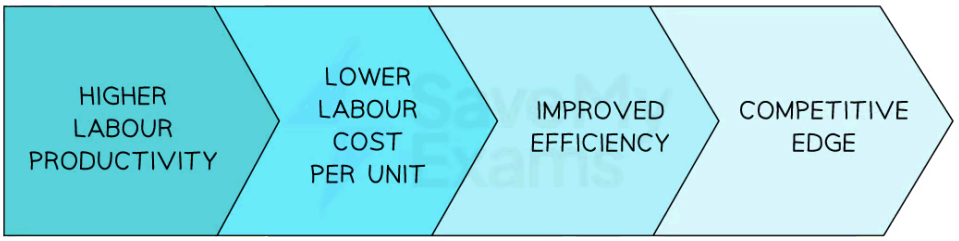
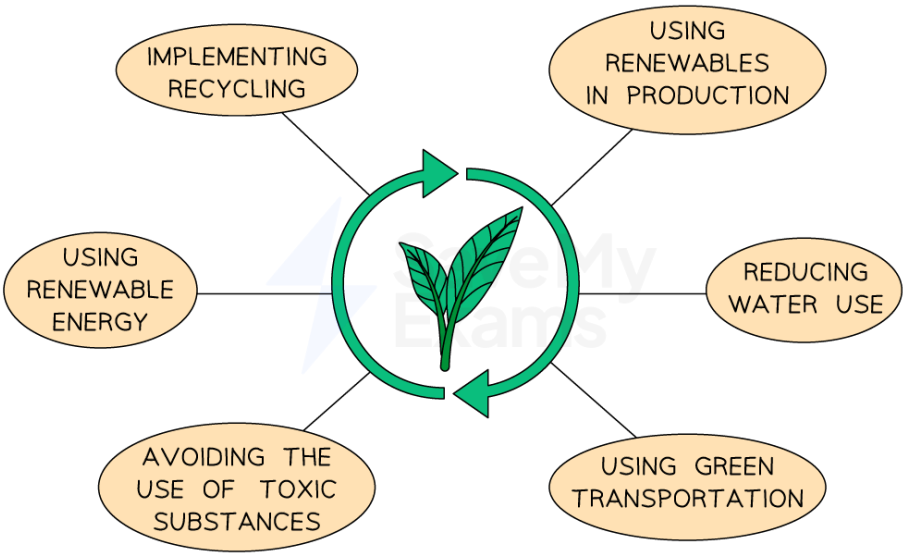


Higher labour productivity improves businesses' competitiveness

- High labour productivity reduces the labour cost per unit, leading to improved efficiency and the potential for a meaningful competitive edge

The impact of improved sustainability

- **Sustainability** means meeting our current needs without compromising the ability of future generations to meet their own needs
- Businesses can contribute to sustainability by changing their operations in a variety of ways



Businesses can contribute to sustainability in a range of ways

Improving sustainability

1. Using renewables in production

- More than 99% of the wood used in **IKEA**'s furniture products is either sourced from sustainable sources or recycled

2. Reducing water use in industrial processes

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- Introduced in 2011, **Levi's** Water<Less programme reduces the use of water in the manufacture of the company's denim products, so far saving over 4.2 billion litres of water in shortage areas

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3. Using green transportation, such as electric vans

- Almost 40% of delivery company **Evri**'s London vehicles are zero-emission and the company operates e-cargo bikes across four UK cities

4. Avoiding the use of toxic substances

- Home accessories brand **Parachute** uses organically-grown textiles, with no harmful chemicals used at any stage of production

5. Using renewable energy, such as solar

- **Unilever** uses 100% renewable electricity across all its factories, offices, **R&D** facilities, data centres, warehouses and distribution centres. They also generate their own solar power at production facilities in 23 countries.

6. Implementing recycling

- **Revive Innovations Ltd** recycles compact discs, turning them into beautiful items of furniture and home accessories

The benefits of improving sustainability

Benefit	Explanation
Meeting customer expectations	▪ Eco-conscious customers prefer sustainable businesses, helping improve brand image and loyalty
Cost savings	▪ Reducing energy use, waste and materials can lower long-term costs
Legal and ethical compliance	▪ Helps meet environmental laws and shows the business is acting responsibly
Competitive advantage	▪ A strong reputation for sustainability can help the business stand out from rivals
Long-term survival	▪ Protects resources and helps the business stay successful in the future

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Capital intensive operations

- **Capital-intensive production** predominately **uses machinery and technology in the production of goods and services**
  - **Large-scale** production of **standardised products** is likely to be capital-intensive
  - Manufacturing in **developed countries** where **labour costs are relatively high** is likely to be capital-intensive
    - E.g. Vehicle manufacturers such as **Ford** use robots and other production technology to manufacture cars, with supervisors overseeing the quality of output

Evaluating capital intensive production

Advantages	Disadvantages
▪ Low-cost production where output is high ▪ Machines are usually consistent and precise ▪ Machines can run without breaks	▪ Significant set-up and maintenance costs ▪ Breakdowns can severely delay production ▪ May not provide flexibility in production

Labour intensive operations

- **Labour-intensive production** predominantly **uses physical labour** in the production of goods and services
  - The **delivery of services** is usually more labour-intensive than manufacturing
  - In countries where **labour costs are low**, such as Bangladesh and Vietnam, labour-intensive production is common
  - **Small-scale production** is likely to be labour-intensive
  - E.g. UK schools are labour-intensive operations as teachers plan and deliver lessons and provide pastoral support

Evaluating labour intensive production

Advantages	Disadvantages
▪ Low-cost production where labour costs are low	▪ Workers may be unreliable and need regular breaks

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▪ Provides opportunities for workers to be creative ▪ Workers are flexible (e.g. they can be retrained)	▪ Incentives may be needed to motivate staff ▪ Training costs can be significant

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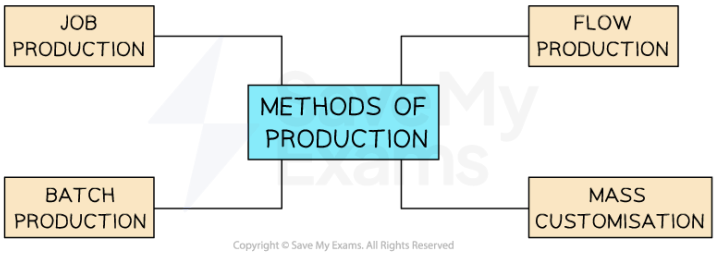
An introduction to production methods

- Production is the **transformation of resources** (e.g. raw materials components and processes) into finished goods or services
  - Goods are **physical products**, such as bicycles and T-shirts
  - Services are **non-physical items** such as hairdressing, tourism and manicures

The four main production methods



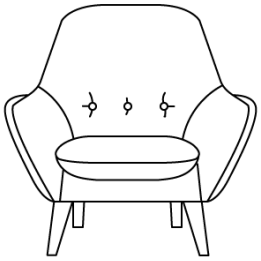
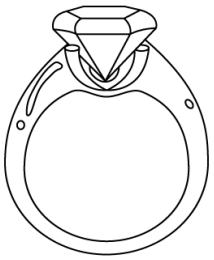
The type of production method used often depends on the nature of the product being made

- The method of production used by a business depends upon a number of factors
  - The **level of output** required to be produced
  - The **nature** of the product
  - Whether the product is **standardised or customised**
  - The level of **automation** used in production

Job production

- Job production is where products are made to meet the **specific requirements of individual customers**
  - Each item is **produced separately** and the production process is **tailored** to the unique specifications of the customer's order
- It is typically used for unique or **specialised** products that are not produced in large quantities
- Job production often requires **skilled labour**, such as craftsmen or technicians, as the manufacturing process may involve intricate tasks or specialised techniques

Examples of job production

Furniture made to order	Tailored clothing	High-end jewellery
Where customers can choose the design, dimensions, materials and finishes	Such as wedding suits, where each garment is made to fit the specific preferences of the individual customer	Especially those with unique designs or personalised engravings
		

- Job production has the potential to add **significant levels of added value** and can therefore be very **profitable**
  - Specialised products made by skilled craftspeople can **command high prices** because they are unique and can be made to **meet customer needs very precisely**
  - This would arguably be lost if other methods of production were to be adopted

Advantages and disadvantages of job production

Advantages	Disadvantages
▪ High customisation – meets unique customer needs	▪ More expensive due to customisation
▪ Flexible – can adapt to changes in demand or trends	▪ Longer lead times – not ideal for urgent orders
▪ High quality – greater attention to detail and control	▪ Complex to manage – requires close coordination with customers
▪ Personalised experience – customers involved in design, increasing loyalty	▪ Low volume – limits ability to achieve economies of scale

Batch production

- Batch production occurs when products are **produced in groups** or batches

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- A certain quantity of products is produced together before moving on to the next batch
- Each batch goes through the **entire production process**, from raw materials to the finished product, before the next batch begins
- Batches are usually of a **standardised size and composition** and follow a certain sequence of operations

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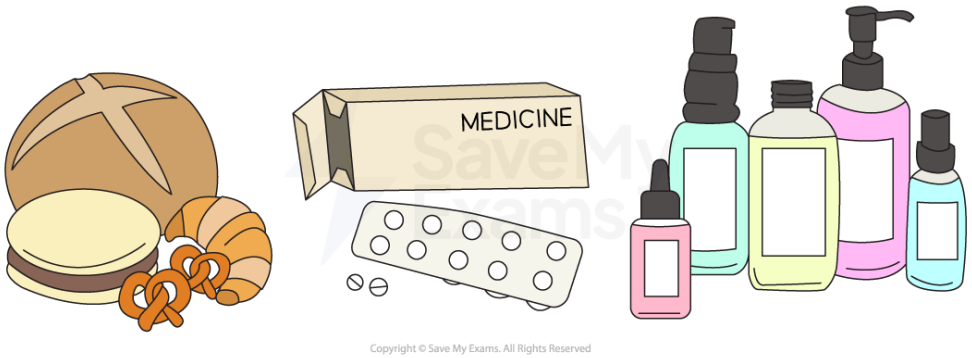
Goods made using batch production



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Commonly used in industries such as pharmaceuticals, beauty products and food processing

- Batch production strikes a **balance between customisation and cost-effectiveness**, making it a suitable production method for industries that deal with **diverse product range**s and **varying customer needs**

Advantages and disadvantages of batch production

Advantages	Disadvantages

▪ Batch production allows manufacturers to **switch between batches** and **cater for varying customer demand** ▪ It can be more **cost-effective** compared to flow production, especially when producing items in smaller quantities ▪ It allows a business to benefit to some extent from **purchasing economies** as larger quantities of stock may be purchased than with job production ▪ **Quality** issues can be identified and **defects** can be rectified within a specific batch before moving on to the next, minimising the impact on the entire production line	▪ Setting up the equipment and configuring the production line for each batch can be **time-consuming** and may result in idle time between batches ▪ Often leads to the **accumulation of inventory**, which requires storage and careful management to avoid wastage ▪ It is **not as adaptable as other production methods,** such as flow production ▪ Rapid changes in product demand or frequent product variations may be difficult to manage ▪ Frequent start-up and shutdown of machinery can put additional **stress on equipment**, requiring regular maintenance and repair to ensure smooth operations

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Mass (flow) production

- Flow production occurs when a product is produced in a **continuous sequence of operations on a production line**
  - It involves the movement of materials or components **through a series of workstations** or machines with each workstation performing a specific task or operation
  - As a product moves along the **production line** it undergoes a series of operations, such as assembly, testing, packaging or quality control until it is completed
- This method is commonly used in industries that produce **high volumes of standardised products** such as automobiles and consumer electronics

Characteristics of flow production

- **Division of labour**
  - Different tasks are allocated to different workstations or machines, allowing workers to specialise in a specific task
- **Standardisation**
  - The manufacture of identical products helps to ensure consistency and the smooth flow of production
- **Continuous movement**

- The product moves continuously from one workstation to another, minimising idle time and maximising productivity
- **High volume**
  - Flow production is suitable for high-volume manufacturing as it enables the efficient production of large quantities of identical or similar products
- **Automation**
  - Flow production often involves the use of machinery and automated equipment to perform repetitive tasks quickly and accurately

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Advantages and disadvantages of flow production

Advantages	Disadvantages

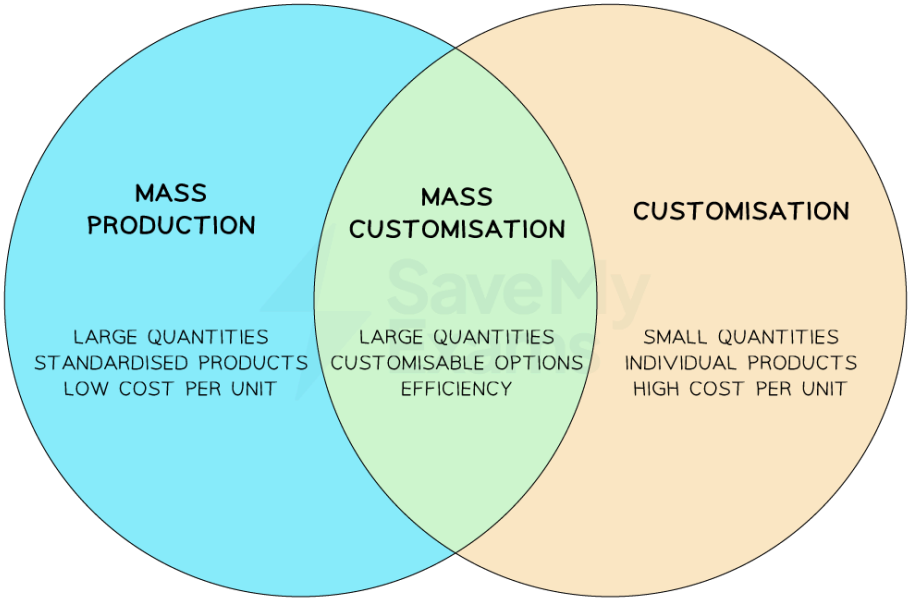
▪ The continuous flow of production **eliminates the need for frequent equipment start-ups and shutdowns**, reducing energy consumption and minimising material waste
▪ **Labour costs may be lower** due to automation and processes that require less skilled workers
▪ It allows for greater **control over product quality** because it is easy to identify and address any defects early on
▪ **It enables fast production,** resulting in **short lead times,** which help companies respond more quickly to market demands
	▪ Implementing flow production systems often requires **significant capital investment** to purchase expensive manufacturing equipment and automation technologies
	▪ It relies on the **reliability and efficiency of equipment and machinery**
	  ▪ If any part of the production line breaks down, it can disrupt the entire process, leading to costly **down time**
	▪ If a defect is detected, it may require the **stoppage of the entire production line,** resulting in substantial losses
	▪ Relies on a **steady supply of components and raw materials**
	  ▪ Any disruption in the supply chain can have a severe impact on production

Examiner Tips and Tricks

When recommending a suitable method of production, carefully consider the needs of the customers. Where the selling price is a key driver of consumer demand, flow production (where **unit costs** are minimised) is likely to be very suitable. Where demand is driven by quality or where customisation is required, job or batch production are likely to be better choices.

Mass customisation

- **Mass production** is usually achieved through flow production and involves the manufacture of **large quantities of standardised products**
  - This results in low costs per unit compared to other methods of production
- **Customisation** is usually associated with job production or small-scale batch production
  - This allows customers to **design products from scratch**
- **Mass customisation** seeks to bridge this gap by offering **flexibility** with **a** range of customisable options within a standardised production process alongside the **low unit costs** typically associated with flow production
  - It uses **technology**, such as computer-aided design (CAD), flexible manufacturing systems and data analytics to efficiently accommodate customer preferences



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A comparison of mass production, customisation and mass customisation

Stages of mass customisation

1. Choice

- Customers are presented with **a range of pre-designed product options**, features or configurations from which they can choose
  - E.g. **Audi** customers can select a specific model, such as the Avant, Saloon, Coupé or Sportback and decide the trim level from Premium, Premium Plus and Prestige

2. Customisation

- Customers have the opportunity to **personalise their chosen product** by selecting specific features, colours, sizes or other configurable elements

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- E.g. **Audi** customers can choose extra features such as driver assistance tools, a towbar or a dashcam

3. Production

- Once the customer's preferences are received, the **manufacturing** process can begin
- The production system is designed to efficiently produce the customised product, often using flexible manufacturing techniques such as **cell production**
  - E.g. At **Audi** each new vehicle takes between six and twelve weeks to manufacture on the highly automated production plant in Germany, with steps involving the use of 3D printing and artificial intelligence

Evaluating mass customisation

- Mass customisation offers several **advantages for both businesses and customers**
  - Customers can obtain **products that match their unique preferences** and requirements and are likely to be **highly satisfied,** which leads to **positive feedback and loyalty**
  - Businesses can **adapt to changing market demands** effectively by offering a **wide range** of product variations **without the expense of full customisation**
  - Using **standardised components and processes** allows businesses to achieve **economies of scale**, reduce stock costs and improve overall production efficiency
  - Mass customisation can **differentiate a business from competitors**, attract new customers and create a perception of **high value** and innovation
- However, mass customisation
  - Usually r**equires heavy capital investment** in technology
  - Tends to lead to **higher variable costs** than for standardised, mass produced items

Problems of changing from one method to another

- A business may want to change its method of production for several reasons
  - It wants to achieve **higher levels of output**
  - It wants to **increase quality**
  - **Labour costs** have increased
  - **Costs of running machinery**, such as power, have increased
- However, it could face a range of **challenges** in making changes to its method of production
  - Increasing the scale of production is likely to require extensive **capital investment** in **machinery**, facilities for the **storage** of inventory and **training** of workers in new processes

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- Using **loans** to finance this investment will increase **fixed costs** and would likely increase the business's level of **gearing**, making it more difficult to borrow more finance in the future
- Changing the method of production places **different demands on workers**
  - Some workers' skills may no longer be required, leading to the need for **redundancies**
  - Others may be **resistant to change**, concerned about the impact on their day-to-day working life and experiences
  - Adjusting to roles that require lower levels of skill or creativity could **demotivate** workers, lowering their **productivity**
- A business needs to ensure that increased demand can be sustained before making significant changes
  - Detailed **market research** and **sales forecasting** that firmly established future demand should be carried out
  - **Outsourcing** may be a more suitable solution in the short-term before capital is invested in **non-current assets** and staff development

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